Fiscal Year:	FY 2006	**Task Last Updated:**	FY 10/19/2005
PI Name:	Kucik, Dennis F. M.D., Ph.D.		
Project Title:	New Automated System for Real-Time Bio-Imaging of a Vascular Cell Function in Microgravity		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Biotechnology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Birmingham	**State:**	AL
Zip Code:	35205-4831	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	01-OBPR-08-B
Start Date:	12/01/2003	**End Date:**	09/30/2007
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 9/30/2007 per PI (11/06)		
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):			
Grant/Contract No.:	NAG9-1561		
Performance Goal No.:			
Performance Goal Text:			
	The purpose of this project is to develop a novel human tissue bioimaging technology applicable to microgravity research. This system will integrate automated bioreactor sampling, Amgel coculture, and real-time bioimaging. This will enable researchers to monitor, for the first time, effects of simulated microgravity on an important vascular function—adhesion between white blood cells and the blood vessel wall-- without interruption of the culture conditions and with essentially no delay between sample extraction and cell-function assay. Technology developed in this project has potential applications for monitoring astronaut immune system function on long-term space flights. We will develop our new, automated system for real-time bio-imaging of effects of microgravity on cell-cell adhesion through completion of the following objectives: 1. Design and develop a system to automatically extract and deliver cell samples from a rotating wall vessel to an assay system at predetermined intervals. 2. Design and develop a small, integrated, real-time digital bio-imaging system to		

Task Description:	produce digital images of cell samples. 3. Integrate into the bio-imaging system an assay of vascular cell function that utilizes the unique Amgel matrix system and sensitive real-time computer tracking of adhesive cell interactions. The system developed for this purpose will be readily adaptable to assay other effects of simulated microgravity on cultured cells. In addition, this technology will be adaptable for in-space astronaut testing. It is critical to NASA's mission for human exploration and development of space to understand the influence of microgravity, on human physiology. Previous work has shown that microgravity can modify host cell-cell and cell-matrix interactions, and thereby exert potentially deleterious effects on human vascular and immune functions and regeneration processes. For simulated microgravity, such as in a rotating wall vessel (RWV), adhesion assays as currently conducted have important limitations. First, to remove a cell sample from the RWV, rotation must be stopped, interrupting the simulated microgravity, not just for the sample, but for all cells in the chamber. Second, adhesion assays can take over an hour to perform when one considers cell preparation, transfer of cells from the microgravity chamber to the assay, and incubation periods. While these delays may be acceptable for general adhesion research, it is important to minimize elapsed time when studying gravitational effects, so that the cells won't have time to recover from microgravity. In this project, we will develop a novel human tissue bioimaging technology applicable to microgravity research to overcome these shortcomings. Lessons learned from development of this Earth-based system will form the basis for a future space-flight culture and assay system to allow NASA to monitor microgravity-inducible physiological events during space flights.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Adhesion of leukocytes to endothelial cells in blood vessels is a key, early step in initiation of inflammation, without which the immune response is ineffective. Much remains to be learned about this process and its molecular basis. Our efforts to better understand the effect of microgravity on leukocyte adhesion will contribute important knowledge to the field of immunology, potentially identifying new targets for therapeutic intervention into diseases involving both immune deficiency and autoimmune mechanisms.
Task Progress:	Objective 1: development of the cell extraction and delivery system. Reliable cell extraction apparatus developed; major improvements in efficient extraction of cells, optimization of valves and control points, miniaturization. Automation of the system is ongoing. Objective 2: development of the real-time bioimaging system. Optical system moved from the optical bench to integration in the self-contained experimental apparatus; mini-microscope tested in actual experiments; tape-storage of data eliminated with development of direct-to-digital video acquisition, streamlining analysis. Objective 3: Integration into the bioimaging system of a cell adhesion assay. New flow chamber developed to facilitate integration into the self-contained experimental apparatus; experiments to test immune-cell function in the integrated system and to determine sensitivity limits are underway.
Bibliography Type:	Description: (Last Updated: 04/12/2018)
Abstracts for Journals and Proceedings	Kucik, D.F.; Rouleau, R.L.; Smith, L.W.; Wu, X.; Gupta, K.B. "Use of an integrated flow-chamber adhesion assay for measuring leukocyte adhesion properties in simulated and actual microgravity. " NASA Cell Science Conference Gravitational and Space Biology Bulletin 19(1) , Nov-2005